

PARADEINE TABLET

MAL19913256AZ

DESCRIPTION:

TABLET

Colour : White
Shape : Round, Flat and Scored
Marking : "P"

EACH TABLET CONTAINS:

Paracetamol 500 mg
Codeine Phosphate 8 mg

PHARMACODYNAMICS:

Codeine Phosphate is a narcotic analgesic that acts in the central nervous system to alter the perception of pain and emotional response to pain. Codeine also has antitussive action by the suppression of the cough reflex by a direct central action, probably in the medulla or pons. Codeine also has antidiarrhoeal action.

Paracetamol has analgesic and antipyretic actions.

PHARMACOKINETICS:

Codeine Phosphate is well absorbed after oral administration and after injection. After an oral dose of 15 mg, peak plasma concentrations of 30 ng/ml are attained within 2 hours. It is widely distributed throughout the body and crosses the placenta. It is metabolised in the liver by O-demethylation to form morphine, N-demethylation to form narcocodeine, and conjugation to form glucuronides and sulphates of both unchanged drug and its metabolites. After an oral dose, about 86% is excreted in the urine in 24 hours; of the excreted material, 40 to 70% is free or conjugated Codeine, 5 to 15% is free or conjugated morphine, 10 to 20% is free or conjugated normorphine; some of the dose is excreted in the bile and trace amounts are found in the faeces; unchanged drug accounts for 6 to 8% of the dose in urine in 24 hours which may be increased to about 10% when the urinary pH is decreased; after an intramuscular dose, 15 to 20% is excreted unchanged in acid urine in 24 hours.

Paracetamol is readily absorbed from the gastrointestinal tract with peak plasma concentrations occurring about 30 minutes to 2 hours after ingestion. It is metabolised in the liver and excreted in the urine mainly as the glycoronide and sulphate conjugates. Less than 5% is excreted as unchanged Paracetamol. The elimination half-life varies from about 1 hour to 4 hours. Plasma protein binding is negligible at usual therapeutic concentrations but increases with increasing concentrations. A minor hydroxylated metabolite which is usually produced in very small amounts by mixed-function oxidases in the liver and which is usually detoxified by conjugation with liver glutathione may accumulate following Paracetamol overdosage and cause liver damage.

INDICATIONS:

Paradeine Tablet is indicated for the relief of painful disorders such as headache, dysmenorrhea, conditions involving musculoskeletal pain, myalgias and neuralgias. It is also indicated as an analgesic and antipyretic in conditions accompanied by discomfort and fever, such as the common cold and viral infections. **Paradeine Tablet** is an effective analgesic after dental work and tooth extractions. Codeine is indicated in patients older than 12 years of age for the treatment of acute moderate pain which is not considered to be relieved by other analgesics such as Paracetamol or Ibuprofen (alone).

RECOMMENDED DOSE:

Adults: 1-2 tablets, 3 to 4 times a day.

Not recommended for children aged 12 to 18 years with compromised respiratory function.

Not to be used for children below the age of 12.

Paediatric population:

- Children aged less than 12 years:
Codeine should not be used in children below the age of 12 years because of the risk of opioid toxicity due to the variable and unpredictable metabolism of Codeine to Morphine.
Paradeine Tablet is contraindicated in children below the age of 12 years for the symptomatic treatment of cold.
- Children aged 12 years to 18 years:
Paradeine Tablet is not recommended for use in children aged 12 years to 18 years with compromised respiratory function.

CONTRAINDICATION:

- Hypersensitivity to Paracetamol or Codeine.
- In children below the age of 12 years for the symptomatic treatment of colds due to an increased risk of developing serious and life-threatening adverse reactions.
- In all paediatric patients (0-18 years of age) who undergo tonsillectomy and/or adenoidectomy for obstructive sleep apnoea syndrome due to increased risk of developing serious and life-threatening adverse reactions.
- In women who are breastfeeding.
- In patients for whom it is known they are CYP2D6 ultra-rapid metabolisers.

WARNING AND PRECAUTIONS:

Do not take more than the recommended dosage or the physician's advice. Codeine may cause drowsiness and it may be habit forming. It should be given with care to patients with impaired kidney or liver function and patients taking other drugs which affect the liver. It should be given with caution to certain patients such as the elderly and to those with hypothyroidism, Addison's disease and prostatic hypertrophy or urethral stricture.

This preparation contains Paracetamol.
Do not take other Paracetamol containing medicines at the same time.

CYP2D6 metabolism

Codeine is metabolised by the liver enzyme CYP2D6 into Morphine, its active metabolite. If a patient has a deficiency or is completely lacking this enzyme an adequate analgesic effect will not be obtained. Estimates indicate that up to 7% of the Caucasian population may have this deficiency. However, if the patient is an extensive or ultra-rapid metaboliser there is an increased risk of developing side

effects of opioid toxicity even at commonly prescribed doses. These patients convert Codeine into Morphine rapidly resulting in higher than expected serum Morphine levels.

General symptoms of opioid toxicity include confusion, somnolence, shallow breathing, small pupils, nausea, vomiting, constipation and lack of appetite. In severe cases this may include symptoms of circulatory and respiratory depression, which may be life-threatening and very rarely fatal. Estimates of prevalence of ultra-rapid metabolisers in different populations are summarised below:

Population	Prevalence %
African/Ethiopian	29%
African American	3.4 to 6.5%
Asian	1.2 to 2.0%
Caucasian	3.6 to 6.5%
Greek	6.0%
Hungarian	1.9%
Northern European	1.0 to 2.0%

Post-operative use in children

There have been reports in the published literature that Codeine given post-operatively in children after tonsillectomy and/or adenoidectomy for obstructive sleep apnoea, led to rare, but life-threatening adverse events including death. All children received doses of Codeine that were within the appropriate dose range; however there was evidence that these children were either ultra-rapid or extensive metabolisers in their ability to metabolise Codeine to Morphine.

Children with compromised respiratory function

Codeine is not recommended for use in children in whom respiratory function might be compromised including neuromuscular disorders, severe cardiac or respiratory conditions, upper respiratory or lung infections, multiple trauma or extensive surgical procedures. These factors may worsen symptoms of Morphine toxicity.

DRUG INTERACTIONS:

None reported

PREGNANCY AND LACTATION:

Pregnancy

Careful consideration should be given before prescribing the product for pregnant patients. Opioid analgesics may depress neonatal respiration and cause withdrawal effects in neonates of dependent mothers.

As a precautionary measure, use of **Paradeine Tablet** should be avoided during the third trimester of pregnancy and during labor.

Breastfeeding

Paradeine Tablet is contraindicated in women during breastfeeding.

At normal therapeutic doses Codeine and its active metabolite may be present in breast milk at very low doses and is unlikely to adversely affect the breast fed infant. However, if the patient is an ultra-rapid metaboliser of CYP2D6, higher levels of the active metabolite, Morphine, may be present in breast milk and on very rare occasions may result in symptoms of opioid toxicity in the infant, which may be fatal.

SIDE EFFECTS/ADVERSE REACTIONS:

Dizziness, nausea and vomiting, gastrointestinal discomfort, lightheadedness, sedation, shortness of breath, allergic reactions, euphoria, constipation, pruritus and respiratory depression.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Codeine Phosphate:

Symptoms of overdosage are respiratory depression, coma, pin-point pupils, cyanosis, weak pulse, hypotension, spasm of GIT and biliary tracts. Death from respiratory failure may occur. Treatment of overdosage consists of induced emesis or gastric lavage, establishment of adequate respiratory exchange through provision of a patent airway and institution of assisted or controlled respiration, and administration of an opioid antagonist, preferably naloxone 10 mcg per kg of body weight or 400 mcg as a single dose, preferably intravenously. Intravenous fluids, vasopressors, and other supportive measures should also be employed as needed. Because the duration of action of the opioid analgesic may exceed that of the antagonist, continued monitoring of the patient is mandatory so that additional antagonist may be administered as needed.

Paracetamol:

Symptoms of acute overdosage include vomiting, anorexia, nausea, epigastric pain. Liver damage, jaundice may follow soon. Profound hypoglycemia and metabolic acidosis might complicate severe poisoning. Gastro-intestinal hemorrhage and acute massive necrosis of the liver and acute tubular necrosis of the kidney may occur. Prompt specific treatment is essential. Any patient who has taken 7.5 g or more of Paracetamol and can be treated within 10 hours of ingestion should be given a specific antidote. Once specific therapy is established, then blood concentrations of Paracetamol can be monitored. Generally treatment is required if the blood concentration is higher than a line (the '200' line) drawn on log/linear paper joining the points 200 mg per liter at 4 hours and 60 mg per liter at 10 hours. There have been suggestions that treatment may be necessary when lower concentrations of Paracetamol are present. Cysteamine is sometimes used by injection as an antidote but acetylcystein or methionine is now preferred. Acetylcystein is given by intravenous infusion in an initial dose of 150 mg per kg over 4 hours and then 100 mg per kg over the next 16 hours. Alternatively, methionine 2.5 g may be given by mouth every 4 hours to a total of 4 doses. Haemoperfusion may be worthwhile if too much time has elapsed since the poisoning to allow use of acetylcystein or methionine. Basic measures that may be required include dextrose and blood infusions. Removal of stomach contents by aspiration and lavage forms an early part of treatment and the administration of Activated Charcoal should be considered.

PACKING/PACK SIZE(S):

Blister pack of 10x10's and 100x10's.

**JAUHI UBAT DARIPADA KANAK-KANAK
KEEP OUT OF REACH OF CHILDREN**

MANUFACTURER/PRODUCT REGISTRATION HOLDER:

DYNAPHARM (M) SDN. BHD. (65683-V)
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Kawasan Perusahaan Perai 3,
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